BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 132 · Mixed Paper 1 and Paper 2 Retrieval

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. A £5-per-ticket program uses total = tickets + 5. What is the correction?

Q2. What does an IDE's run-time environment let you do?
A. Repair physical CPU faults
B. Rearrange HDD fragments
C. Manage network user accounts
D. Run and test code

Q3. “An IPv4 address is always six pairs of hexadecimal digits separated by colons.” Correct this. Explain why.

Q4. An IDE finds a missing bracket. Once fixed, the program runs but gives the wrong total. Why did the IDE not prove the logic correct? What should you do next?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Synoptic retrieval — Recall previously learned Paper 1 and Paper 2 material in mixed order.

Choose the method — Decide whether a task needs an explanation, calculation, trace or evaluation.

Link systems and coding — Variables, memory and data representation affect the way programs execute.

Identify gaps — Use wrong answers to choose the next targeted practice rather than repeat every topic.

New learning

Synoptic retrieval: Recall previously learned Paper 1 and Paper 2 material in mixed order.

Choose the method: Decide whether a task needs an explanation, calculation, trace or evaluation.

Link systems and coding: Variables, memory and data representation affect the way programs execute.

Identify gaps: Use wrong answers to choose the next targeted practice rather than repeat every topic.

Worked example

A monochrome 100 × 50 bitmap uses 1 bit/pixel. Calculate its uncompressed file size in bytes.

1. Compute pixels: 100×50=5000.
2. Multiply by one bit per pixel: 5000 bits.
3. Convert 5000 ÷ 8=625 bytes.
4. State assumptions: ignoring metadata and compression.

We Do · Practise with guidance

Guided 1

A mono sound clip lasts 5 seconds at 8000 samples/s and 8 bits/sample. Find bytes.

Worked method: 40,000 bytes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the MOD calculation that finds leftover bits.

```
bits = 203
leftover = ...
print(leftover)
```

write

Question 1

Calculate the denary value of unsigned binary 10101010. [3 marks]

Question 2

Explain why binary search requires a sorted list. [3 marks]

Question 3

Explain one advantage and one limitation of virtual memory. [4 marks]

Exit Ticket · Answer before checking

Image file size in bytes equals width×height×bit depth without any conversion.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Use total = tickets * 5; addition does not calculate £5 for each ticket.

Q2. D — Run and test code

Q3. IPv4 supports addressing and routing. It is usually written as four decimal numbers separated by dots.

Q4. The IDE found a syntax error, not a logic error. Trace the calculation. Test inputs with known totals and compare results.

I Do / We Do pages · Guided practice

Q1. 40,000 bytes.

You Do pages · Code First

```
bits = 203
leftover = bits MOD 8
print(leftover)
```

Expected result: 3

You Do pages · Independent questions

Q1. 128+32+8+2 = 170.

Q2. Midpoint comparisons only allow safe elimination of half the list when values are ordered.

Q3. Lets the OS accommodate memory demand beyond physical RAM; uses slower secondary storage, so paging can reduce performance.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: